

BIODIVERSITY AND CONSERVATION

1. Match the following–

(a) Ants	(i) 20,000 species
(b) Beetles	(ii) >28,000 species
(c) Fishes	(iii) >3,00,000 species
(d) Orchids	(iv) >20,000 species

 - (1) a (iv), b (ii), c (iii), d (i)
 - (2) a (ii), b (iii), c (iv), d (i)
 - (3) a (iv), b (iii), c (ii), d (i)
 - (4) a (iii), b (ii), c (i), d (iv)
2. In biosphere, diversity (heterogeneity) exist at –
 - (1) Species level
 - (2) Genetic level
 - (3) Ecosystem level
 - (4) All the above
3. Term "**biodiversity**" was popularised by–

(1) Edward Wilson	(2) Humboldt
(3) Tilman	(4) Paul Ehrlich
4. In India, how many genetically different strains of rice and mango varieties are present –
 - (1) <50,000 and 1,0000 respectively
 - (2) 1000 and 50000 respectively
 - (3) >50.000 and 1.000 respectively
 - (4) >50,000 and 5,000 respectively
5. Find out incorrect statement –
 - (1) Biodiversity exists at all levels of biological organisation
 - (2) A single species cannot show more diversity at genetic level
 - (3) Western ghats have a greater amphibian species diversity
 - (4) India has greater ecological diversity than Scandinavian country.
6. According to IUCN (2004), how many plant and animal species have been described so far –

(1) <1.5 billion	(2) >1.5 million
(3) 7.1 million	(4) 7.1 billion
7. Which method is thought to be best for estimation of microbial biodiversity –
 - (1) Bio statistical method
 - (2) Bio chemical or molecular method
 - (3) Paleobotanical method
 - (4) Culture method
8. What is the contribution of India in global species diversity –

(1) 2.4 %	(2) 12 %
(3) 8.1 %	(4) 7.1 %
9. Find out the right one –
 - (1) India has 7.1% species diversity
 - (2) India is one of the 22 mega diversity countries of world
 - (3) According to Robert May's global estimate only 22% of the total species have been recorded so far
 - (4) According to Robert May's estimation, more than 3,00,000 plant species yet to be discovered in India
10. Pattern of Biodiversity depends upon –
 - (1) Latitudinal gradient
 - (2) Altitudinal gradient
 - (3) Species - area relationship
 - (4) All of these
11. Which of the following is not the reason of great biodiversity in tropics ?
 - (1) Undisturbed climate for millions of years
 - (2) Less seasonal variations
 - (3) More nutritive soil
 - (4) More solar energy
12. Which statement is incorrect –
 - (1) There is increase in biodiversity from pole to equator
 - (2) There is increase in biodiversity from low altitude to high altitude
 - (3) There is directly proportional relationship between area & biodiversity
 - (4) Tropics has less seasonal, relatively more constant and predictable environment.

- 13.** In species area relationship, on a logarithmic scale. The relationship is –
- (1) Rectangular hyperbola
 - (2) Rectangular parabola
 - (3) Straight line
 - (4) Sigmoid
- 14.** In equation $\log S = \log C + Z \log A$, what is Z –
- (1) Species richness
 - (2) Area
 - (3) Regression coefficient
 - (4) Y-intercept
- 15.** For frugivorous birds and mammals in tropical forest of different continents, the slope(Z) is found to be :-
- (1) 0.1 - 0.2
 - (2) 0.6 - 1.2
 - (3) 1.15
 - (4) 11.5
- 16.** More species in community, tends to more stability than communities with less species'. It was supported by–
- (1) David Tilman
 - (2) Paul Ehrlich
 - (3) Humboldt
 - (4) Tansley
- 17.** "Rivet popper hypothesis" was proposed by –
- (1) Tilman
 - (2) Hombolat
 - (3) Paul Ehrlich
 - (4) Mayer
- 18.** Select the wrong match for extinction of species as per the IUCN red list 2004 –
- (1) Vertebrates – 338
 - (2) Invertebrates - 359
 - (3) Plants - 87
 - (4) Prokaryotes - 2001
- 19.** Careful analysis of records shows that extinction across taxa are not random, some groups like _____ appear to be more vulnerable to extinction.
- (1) Reptiles
 - (2) Mammals
 - (3) Aves
 - (4) Amphibians
- 20.** According to ecologists warn that if the present trends continue. Nearly half of the species on earth might be wiped out within how many years –
- (1) 100
 - (2) 1000
 - (3) 50
 - (4) 500
- 21.** In general, the loss of biodiversity in a region may not lead to –
- (1) Decline in plant production
 - (2) Lowered resistance to environmental perturbations
 - (3) Constant pest and disease cycles
 - (4) Increased variability in certain ecosystem processes such as plant productivity
- 22.** From 'The Evil Quartet' which of the following is most important cause of biodiversity loss –
- (1) Habitat loss and fragmentation
 - (2) Over - exploitation
 - (3) Alien species invasion
 - (4) Co-extinctions
- 23.** The most dramatic examples of habitat loss come from tropical rain forests. Once covering more than 14 percent of earth's land surface and now cover no more than –
- (1) 5%
 - (2) 6%
 - (3) 10%
 - (4) 13%
- 24.** The Amazon rain forest 'lungs of Planet' harbouring probably millions of species is being cut & cleared for which purpose –
- (1) For cultivation of soyabean
 - (2) For conversion to grasslands for raising beef cattle
 - (3) For cultivation of Medicinal plants
 - (4) Both (1) and (2)
- 25.** When large habitats are broken up into small fragments due to human activities, which of the following get badly affected –
- (1) Mammals and birds requiring large territories
 - (2) Animals with migratory habitats
 - (3) Animals with large body size
 - (4) Both (1) and (2)

- 26.** Alien species invasion is one of the cause of Biodiversity loss. Introduction of Nile perch in victoria lake lead to extinction of more than 200 species of –
(1) Cichlid fish (2) Gambusia
(3) Salmon fish (4) Cat fish
- 27.** The recent illegal introduction of which African fish becomes severe cause of threatening of indigenous cat fishes –
(1) Gambusia (2) Labeo
(3) *Clarias gariepinus* (4) Dog fish
- 28.** Find out the following w.r.t. economic importance of biodiversity –
(1) Production of oxygen (2) Pollination
(3) Medicinal utility (4) All
- 29.** When we conserve and protect the whole ecosystem, its biodiversity at all levels is called –
(1) In situ conservation (2) Ex -situ conservation
(3) On site conservation (4) Both (1) and (3)
- 30.** Which among the following is the criterion for determination of 'Biodiversity hot spot' –
(1) Very high level of species richriess
(2) High degree of habitat loss and fragmentation
(3) High degree of Endemism
(4) All of the above
- 31.** Find out the odd one with respect to 'Biodiversity hot spots' –
(1) Western Ghats & Srilanka
(2) Indo - Burma
(3) Himalaya
(4) Gangatic plains
- 32.** Find out the wrong match –
(1) Bioshpere reserves – 14
(2) National parks – 85
(3) Wild life sanctuaries – 448
(4) Indian Biodiversity hot spot – 3
- 33.** 'Sacred groves is also one of the important mean of Biodiversity conservation. In respect of this find out the odd one –
(1) Khasi and Jaintia – Meghalaya
(2) Aravalli hills – Rajasthan
(3) Sarguja, Chanda and Bastar – Mizoram
(4) Western Ghat – Maharashtra